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ATOMIZER AND AEROSOL GENERATING APPARATUS

Abstract

An atomizer includes: a liquid cup provided with a mist outlet; an atomization assembly including an atomization tube, a liquid guide part, and a heating part disposed inside the liquid guide part, the atomization tube being sleeved on an outer peripheral side of the liquid guide part and having a first end connected with the mist outlet and a second end provided with a liquid inlet; and a sealing member including a support part and a protrusion. The support part abuts against an inner wall of the liquid cup. The support, the liquid cup and the atomization tube forms a liquid storage tank. The protrusion extends into the liquid guide part from an end of the liquid guide part away from the mist outlet. The support part is provided with at least one ventilation passage for connecting the liquid storage tank with atmosphere.

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Background/Summary

RELATED APPLICATION

[0001] This application is a Paris Convention, which claims the benefit of priority of Chinese [0002] patent application Ser. No. 20/242,0451396.0 filed on Mar. 7, 2024. The contents of the above application is all incorporated by reference as if fully set forth herein in its entirety.

FIELD AND BACKGROUND OF THE INVENTION

[0003] The present disclosure relates to atomization technologies, and more particularly relates to an atomizer and an aerosol generating apparatus.

[0004] An atomizer is a part of an aerosol generating apparatus for generating aerosol, and therefore is provided with an atomization assembly for atomizing liquid and mixing the atomized liquid with gas to generate aerosol for use by a user. For this purpose, a liquid storage tank is provided in a liquid cup, and may supply liquid to the atomization assembly for atomization.

[0005] To avoid leakage of liquid from the liquid storage tank, the liquid storage tank is generally a closed structure. Thus, when the liquid in the liquid storage tank is consumed, the volume of the gas in the liquid storage tank increases, so that the gas pressure is reduced. The reduced gas pressure may prevent liquid from flowing out to the heating part through a liquid guide. Generally, the atomizer may wait until the gas pressure drops to a certain extent to form a sufficiently strong negative pressure, so that external gas is directly sucked into the liquid storage tank through the liquid guide to balance the negative pressure.

[0006] However, this requires a great negative pressure. Usually, before the external gas is sucked into the liquid storage tank, the liquid discharge speed has been seriously affected, so that the heating part of an atomizer having a great mist discharge flow rate could come into contact with less liquid. This may result in dry burning and burn-out of the heating part and the liquid guide. Thus, there is a need for an atomizer with a stronger liquid supply capacity.

SUMMARY OF THE INVENTION

[0007] According to one or more embodiments of the present disclosure, an atomizer includes: a liquid cup having an end provided with a mist outlet; an atomization assembly comprising an atomization tube, a liquid guide part, and a heating part, the heating part being disposed in the liquid guide part, the atomization tube being sleeved on an outer peripheral side of the liquid guide part, and having a first end connected with the mist outlet and a second end provided with a liquid inlet; and a sealing member comprising a support part and a protrusion, the support part abutting against an inner wall of the liquid cup, the support, the liquid cup and the atomization tube forming a liquid storage tank connected with the liquid guide part through the liquid inlet. The protrusion extends into the liquid guide part from an end of the liquid guide part away from the mist outlet, so that the protrusion, the atomization tube and the mist outlet form a mist outlet passage, and the support part is provided with at least one ventilation passage for connecting the liquid storage tank with atmosphere.

[0008] According to one or more embodiments of the present disclosure, an aerosol generating apparatus includes an atomizer as described above and a body.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0009] FIG. **1** is a schematic cross-sectional view of an example of an atomizer according to one or more embodiments of the present disclosure.

[0010] FIG. **2** is an enlarged view of part C in FIG. **1**.

[0011] FIG. **3** is a schematic cross-sectional view of the atomizer as shown in FIG. **1** from another perspective.

[0012] FIG. **4** is an enlarged view of part D in FIG. **3**.

[0013] FIG. **5** is a schematic structural diagram of a sealing member according to one or more embodiments of the present disclosure.

[0014] FIG. **6** is a schematic structural diagram of the sealing member as shown in FIG. **5** from another perspective.

[0015] FIG. **7** is a schematic cross-sectional view of another example of an atomizer according to one or more embodiments of the present disclosure.

[0016] FIG. **8** is an enlarged view of part E in FIG. **7**.

[0017] FIG. **9** is a schematic cross-sectional view of the atomizer as shown in FIG. **7** from another perspective.

[0018] FIG. **10** is a schematic exploded view of the atomizer as shown in FIG. **1**.

[0019] FIG. **11** is an enlarged view of part F in FIG. **10**.

[0020] FIG. **12** is a structural diagram of the atomizer as shown in FIG. **1** in an intermediate state during the assembly process based on an example.

[0021] FIG. **13** is a structural diagram of the atomizer as shown in FIG. **1** in an intermediate state during the assembly process based on another example.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

[0022] Some embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings. The embodiments are described for illustrative purposes only and are not intended to limit the present disclosure.

[0023] The atomizer according to one or more embodiments of the present disclosure may be applied to an aerosol generating apparatus. The aerosol generating apparatus may include the atomizer and a body, and a unit such as a control unit and a power supply unit may be provided in the body. When the atomizer is connected to the body of the aerosol generating apparatus, the power supply unit in the body supplies power to the atomizer, so that the power supply unit can supply power to the atomization assembly of the atomizer to generate heat. The specific manner in which the atomization assembly generates heat is determined by the type of the atomization assembly. For example, when the atomization assembly includes a structure such as an electric heating wire, current is directly supplied to the atomization assembly, thereby heating the electric heating wire. When the atomization assembly is a conductor, a changing magnetic field may be supplied to the atomization assembly to excite the atomization assembly to generate a vortex current to generate heat. The control assembly in the body of the aerosol generating apparatus may also be provided with a breath detector. When the atomizer is connected to the body, the breath detector may be connected with a mist outlet tube so that the breath detector can detect the gas flow in the mist outlet tube, thereby controlling the power supply unit to supply to the atomization assembly electric power suitable for the gas flow in the mist outlet tube.

[0024] As shown in FIG. **1**, an atomizer according to one or more embodiments of the present disclosure includes a liquid cup **100**, an atomization assembly **200**, and a sealing member **310**. An end of the liquid cup **100** is provided with a mist outlet **121**. In one or more embodiments having a mist outlet tube **120**, the mist outlet **121** is provided at an end of the mist outlet tube **120** (see FIG. **1**). In one or more embodiments without a mist outlet tube **120**, the mist outlet **121** may be served by an installation opening on the housing **110**. The atomization assembly **200** includes an atomization tube **210**, a liquid guide, and a heating part **240**. The heating part **240** is disposed in the

liquid guide which is for guiding liquid to the heating part **240**. The atomization tube **210** is sleeved on the outer peripheral side of the liquid guide, and is provided with a liquid inlet **211**. The atomization tube **210** is connected with the mist outlet **121**. The sealing member **310** includes a support part **314** and a protrusion **312**. The support part **314** abuts against the inner wall of the liquid cup **100**, and the support part **314**, the liquid cup **100** and the atomization tube **210** form the liquid storage tank **140**. The liquid storage tank **140** is connected with the liquid guide through the liquid inlet **211**. The protrusion **312** extends through the support part **314** and extends into the liquid guide from the end of the liquid guide away from the mist outlet **121**, so that the protrusion **312**, the atomization tube **210** and the mist outlet **121** are connected to form a mist outlet passage. The support part is provided with at least one ventilation passage for connecting the liquid storage tank with external atmosphere. According to the atomizer provided by one or more embodiments of the present disclosure, the support part is provided with at least one ventilation passage for connecting the liquid storage tank and the external atmosphere, so that the liquid supply capacity of the atomizer is better. In particular, the ventilation passage connecting the liquid storage tank and the external atmosphere provides a new gas inlet passage for the gas. The gas can more easily enter the liquid storage tank through the ventilation passage without passing through the liquid guide, so that when the negative pressure in the liquid storage tank is small, the gas is guided into the liquid storage tank through the ventilation passage, thereby enabling the liquid storage tank to be replenished with the external gas in time, and improving the liquid supply capability of the atomizer.

[0025] The protrusion **312** at least partially fits the inner wall of the liquid guide and a first gap **312c** is formed at the fit position. The first gap **312c** may be formed by a clearance fit between the liquid guide and the protrusion **312**. However, a gas flow passage may be formed between the liquid guide and the protrusion **312** even when the liquid guide and the protrusion **312** abut against each other. The reason is that the materials of the liquid guide and the protrusion **312** are different, so that the attraction of each of the liquid guide and the protrusion **312** to the liquid is different. When the gas needs to pass through the fit position of the liquid guide and the protrusion **312**, since the liquid is subjected to different attractions respectively from the protrusion **312** and the liquid guide, it is likely to be detached from the liquid guide or the protrusion **312** under the influence of the gas, so that a gas passage without liquid is formed in a thin layer on the inner peripheral side of the liquid guide attached to the protrusion **312**, that is, the first gap **312c**. The liquid guide is at least partially spaced apart from the support part **314** to form a second gap. The first gap **312c** is connected with the second gap to form a gas inlet passage connecting the mist outlet passage and the liquid storage tank **140**. As shown FIG. 2, the gas may be drawn from the gas passage in the middle of the atomization assembly **200** or the protrusion **312** through gas inlet passage and guided into the liquid storage tank **140**. The reason for this design is that the liquid in the liquid storage tank is gradually consumed with the use of the user, so that the space in the liquid storage tank becomes larger. If there is no new gas replenished into the liquid storage tank, the gas volume in the liquid storage tank becomes greater, and the gas pressure decreases. When the gas pressure in the liquid storage tank **140** is less than atmospheric pressure, the gas in the liquid storage tank **140** acts to draw the liquid back into the liquid storage tank, preventing the liquid from being guided to the heating part **240** through the liquid guide. Thus, insufficient liquid guided to the heating part **240** may cause the heating part **240** to overheat and burn out the liquid guide. Therefore, the first gap **312c** and the second gap connected with each other are provided, so that the gas can be replenished in the channel liquid storage tank **140** in time, thereby preventing the negative pressure and improving the liquid supply capacity of the atomizer.

[0026] As shown in FIG. 5, in one or more embodiments, an end face of the liquid guide facing the support part **314** abuts against the support part **314**. In this way, the support part **314** may provide support for the liquid guide and improve the stability of the liquid guide. A gas release groove **314a** of the ventilation passage is provided on a side of the support part **314** facing the liquid guide to

form a second gap, that is, the ventilation passage at least includes the gas release groove **314a** and is connected with the first gap **312c**. This also ensures the formation of the second gap and the liquid supply capacity of the aerosol generating apparatus.

[0027] As shown in FIG. 4, in one or more embodiments, the gas release groove **314a** extends in the radial direction of the protrusion **312**. The atomization tube **210** has a projection on the sealing member, and the gas release groove **314a** protrudes out of the projection in the radial direction of the protrusion **312**. As can be seen in FIG. 4, a projection of the protrusion **312** in an axial direction on the support part **314** does not cover the gas release groove **314a**, which ensures that the gas release groove **314a** has an enough length and there is an enough vent length enable the gas to enter the liquid storage tank **140** smoothly.

[0028] As shown in FIG. 5, in one or more embodiments, there are multiple gas release grooves **314a**, which are arranged in the circumferential direction of the protrusion **312** at intervals. In this way, during the use of the atomizer, although the user may use the atomizer at various angles, there is at least one of the gas release grooves **314a** at each angle to ensure the stability of the use of the atomizer. The effect of gas release is prevented from being poor at some angles. However, as shown in FIG. 7, in one or more embodiments, the gas release groove **314a** may not be provided, and the support part **314** and the end face of the liquid guide close to the support part **314** may be spaced apart from each other, thereby forming the second gap.

[0029] As shown in FIGS. 1 and 2, in one or more embodiments, the liquid guide includes a first liquid guide **220** and a second liquid guide **230**. The atomization tube **210** is sleeved on the outer peripheral side of the first liquid guide **220**. The first liquid guide **220** is sleeved on the outer peripheral side of the second guide **230**. The heating part **240** is disposed in the second liquid guide **230**. The radially outer peripheral side of the protrusion **312** fits the radially inner peripheral side of the first liquid guide **220** to form the first gap **312c**. The end face of the protrusion **312** away from the support part **314** fits the end face of the second liquid guide **230** to form a third gap, which is connected with the first gap **312c**. The principle in which the protrusion **312** fits the first liquid guide **220** to form the first gap **312c** is the same as that in which the protrusion **312** fits the liquid guide to form the first gap **312c**, which is not repeated. The principle of forming the third gap may be the same as that of forming the first gap **312c**. However, a gas groove extending radially along the second protrusion **312** may be provided on the end face of the protrusion **312** facing the second liquid guide **230**, and the gas groove connects the gas passage in the protrusion **312** with the first gap **312c**, so that the third gap is formed by the gas groove. The arrangement of the third gap may improve the gas inlet effect of the gas inlet passage into the liquid storage tank **140**.

[0030] As shown in FIG. 1, in one or more embodiments, the sealing member **310** further includes a first extension part **315**. The first extension part **315** is provided on the peripheral side of the support part **314** and extends away from the liquid storage tank **140**. This is equivalent to providing the support part **314** with a skirt. The outer peripheral side of the first extension part **315** abuts against the inner wall of the liquid cup **100**. This is equivalent to increasing the contact area between the support part **314** and the inner wall of the liquid cup **100**, so that when the liquid penetrates into the fit clearance between the support part **314** and the liquid cup **100** and the support part **314** is broken through, the first extension part **315** further generates a sealing effect, thereby preventing the leakage of the liquid and improving the sealing effect.

[0031] As shown in FIG. 5, in one or more embodiments, the outer peripheral side of the first extension part **315** is provided with a convex rib **315a**, which extends along the circumferential direction of the first extension part **315** and surrounds the first extension part **315**. Thus, when the outer peripheral side of the first extension part **315** abuts against the inner wall of the liquid cup **100**, the convex rib **315a** also abuts against the inner wall of the liquid cup **100**. However, since the convex rib **315a** protrudes out of the first extension part **315** in the radial direction of the first extension part **315**, the pressure between the convex rib **315a** and the liquid cup **100** may be greater than the pressure between another portion of the first extension part **315** and the liquid cup **100**.

This allows the convex rib **315a** to better seal the fit clearance between the first extension part **315** and the cup **100**, thereby improving the sealing effect.

[0032] As shown in FIG. **8**, in one or more embodiments, the liquid guide includes the first liquid guide **220** and the second liquid guide **230**. The atomization tube **210** is sleeved on the outer peripheral side of the first liquid guide **220**. The first liquid guide **220** is sleeved on the outer peripheral side of the second guide **230**. The heating part **240** is disposed in the second liquid guide **230**. The radially outer peripheral side of the protrusion **312** fits the radially inner peripheral side of the first liquid guide **220** to form the first gap **312c**. The end face of the protrusion **312** away from the support part **314** fits the end face of the second liquid guide **230** to form a third gap. The sealing member **310** also includes a heat insulation sleeve **313** which is sleeved on the end of the protrusion **312** close to the second liquid guide **230**. As shown in FIG. **7** or FIG. **8**, in one or more embodiments, the protrusion **312** and the reinforcing sheet **316** are integrated. Thus, the protrusion **312** has the same material as the reinforcing sheet **316**, and may have a plastic having a low heat resistance. Therefore, the heat insulation sleeve **313** is provided to better separate the protrusion **312** from the heating part, so that the protrusion **312** may be protected. As shown in FIG. **9**, the heat insulation sleeve **313** may also extend toward the base **320** in the axial direction of the protrusion **312**. In this way, a first lead channel **312a** and a second lead channel **312b** may be provided in the heat insulation sleeve **313**. It will be readily appreciated that, as shown in FIG. **1**, the end of the protrusion **312** is not provided with the heat insulation sleeve **313**, while as shown in FIG. **8**, the end of the protrusion **312** close to the second liquid guide **230** is sleeved with the heat insulation sleeve **313**.

[0033] As shown in FIG. **1** or FIG. **10**, in one or more embodiments, the atomizer further includes a liquid injection cover **600**. The wall of the liquid cup **100** is further provided with a liquid injection hole connected with the liquid storage tank **140**, and the liquid injection cover **600** covers the liquid injection hole. Thus, when there is not sufficient liquid in the liquid storage tank **140**, the liquid injection cover **600** may be opened to fill the liquid storage tank **140**, which allows the atomizer to be reused.

[0034] As shown in FIG. **1**, in one or more embodiments, the liquid injection hole is spaced apart from the second gap in the axial direction of the atomization assembly **200**. In this way, when the liquid is filled, it is not easy for the liquid to directly flush into the second gap, which causes the liquid to flow from the second gap into the first gap **312c**, and finally flow into the gas passage in the atomization assembly **200**, thereby causing leakage of the liquid.

[0035] As shown in FIG. **1**, according to one or more embodiments of the present disclosure, an atomizer includes the liquid cup **100**, the atomization assembly **200**, and the base assembly **300**. The liquid cup **100** may include a housing **110** and a mist outlet tube **120** having one end provided with a mist outlet **121**. The end of the mist outlet tube **120** provided with the mist outlet **121** is connected to the housing. The atomization assembly may include a first end and a second end opposite to the first end. The first end of the atomization assembly is connected to an end of the mist outlet tube **120** away from the mist outlet **121**. The base assembly includes the protrusion **312** and the support part connected to the protrusion **312**. The protrusion **312** is connected to the second end of the atomization assembly **200**, and an end of the support part away from the protrusion is connected to an end of the housing away from the mist outlet **121**. The protrusion **312** may have a material such as silica gel, rubber, or the like. In one or more embodiments, the protrusion **312** may also be provided with a gas flow passage **3122** connected with outside. Thus, the outside and the gas path of the atomization assembly can be connected.

[0036] Specifically, as shown in FIG. **12**, in one or more embodiments, an installation chamber **130** is provided in the housing **110**. The housing **110** is further provided with a cup mouth and an installation opening both connected with the installation chamber **130**, and the mist outlet tube **120** is provided in the installation chamber **130**. An end of the mist outlet tube **120** is installed to the installation opening and provided with the mist outlet **121**, and the other end of the mist outlet tube

120 extends toward the cup mouth. In practice, in one or more embodiments without the mist outlet tube **120**, the installation opening may directly serve as the mist outlet **121**. As shown in FIG. 1, the edge of the installation opening and the edge of the mist outlet **121** are connected to each other, so that the mist outlet tube **120** and the housing **110** may be integrated, or the mist outlet tube **120** may be separately formed and then connected to the housing **110** by means of bonding or fastening. An end of the atomization assembly **200** is connected to the mist outlet tube **120** by insertion. As shown in FIG. 1, the atomization tube **210** of the atomization assembly **200** may be inserted into the mist outlet tube **120**. However, the mist outlet tube **120** may be inserted into the atomization tube **210** of the atomization assembly **200** or into the liquid guide of the atomization assembly **200**. An end of the base assembly **300** close to the mist outlet tube **120** is connected to an end of the atomization assembly **200** away from the mist outlet tube **120** by insertion. Similarly, as shown in FIG. 1, the base assembly **300** may be provided with a protrusion **312**. The protrusion **312** is inserted into the liquid guide or the atomization tube **210** of the atomization assembly **200** is inserted into the base assembly **300**. The base assembly **300** covers the cup mouth so that the structure in the installation chamber **130** can be enclosed.

[0037] As shown in FIG. 1, it can be seen that, on the one hand, the atomization assembly **200** does not need to be installed in the mist outlet tube **120** (generally, the atomization assembly **200** is installed in the mist outlet tube **120**), but is simply connected to the mist outlet tube **120** by insertion, so that it is not necessary to install the atomization assembly **200** in the mist outlet tube **120** with a narrow space. It is difficult to install the atomization component **200** in a preset position inside the mist outlet tube **120**, which makes it difficult to assemble the atomization assembly **200**, thereby causing low assembling efficiency and poor consistency. On the other hand, the length of the mist outlet tube **120** is shortened. This is because if the atomization assembly **200** is installed in the mist outlet tube **120**, the end of the mist outlet tube **120** away from the mist outlet **121** has to extend to the base assembly **300**. This makes it simpler to assemble even if the mist outlet tube **120** and the housing **110** are separated and need to be assembled. This is because if the mist outlet tube **120** is longer, the installation deviation of the end of the mist outlet **120** away from the mist outlet **121** will be amplified more when the installation deviation of the end of the mist outlet tube **120** at the installation opening is the same. In addition, since the atomization assembly **200** is connected to each of the mist outlet tube **120** and the base assembly **300** by insertion, the atomization assembly **200** and the base assembly **300** may be assembled first, and then the atomization assembly **200** is connected to the mist outlet tube **120** by insertion, and then the base assembly **300** is connected to the atomization assembly **200** by insertion to complete installation. The assembly is modular and more convenient (for details, reference may be made to the contents of the assembling method of the atomizer hereinafter).

[0038] As shown in FIG. 1, in one or more embodiments, an atomization assembly **200** includes the atomization tube **210**, the liquid guide, and the heating part **240**. The heating part **240** is disposed on the inner side of the liquid guide, and the liquid guide includes a first liquid guide **220** and a second liquid guide **230**, that is, as shown in FIG. 3, the heating part **240** may be disposed on the inner peripheral side of the second liquid guide **230**. In another one or more embodiments, the second liquid guide **230** may surround the heating part **240**, or when the second liquid guide **230** is not provided and only the first liquid guide **220** is provided, the atomization assembly **200** may be disposed on the inner peripheral side of the first liquid guide **220** or surrounded by the first liquid guide **220**. The atomization tube **210** is sleeved on the outer peripheral side of the liquid guide. The atomization tube **210** is connected to an end of the mist outlet tube **120** away from the mist outlet **121** by insertion so as to be connected with the mist outlet tube **120**. An end of the base assembly **300** close to the mist outlet tube **120** is connected to an end of the liquid guide away from the mist outlet tube **120** and fits the inner peripheral side of the end of the liquid guide away from the mist outlet tube **120**. This may improve the sealing of the atomizer, that is, prevent the leakage of liquid from the liquid storage tank **140** through the atomization assembly **200**. In particular, the liquid is

guided from the liquid storage tank **140** and dispersed throughout the atomization assembly **200** by the liquid guide through capillary action. Since being disposed in the liquid guide, the heating part **240** is in contact with the liquid on the liquid guide, and when the heating part **240** heats up, the liquid is atomized and mixed with gas to generate aerosol which is finally discharged from the mist outlet **121** for use by a user. The distribution of the liquid in the liquid guide is not completely uniform, since under the action of gravity the liquid tends to accumulate towards the direction in which the gravity is directed. Thus, if the atomizer is placed upright as shown in FIG. **3**, in one or more embodiments, the liquid tends to accumulate in the direction of the liquid guide close to the support part **314**. Therefore, it is easy for the liquid accumulated at the position in the liquid guide close to the support part **314** to break through the limit of the liquid guide and flows out of the liquid guide. However, since an end of the base assembly **300** close to the mist outlet tube **120** is connected to an end of the liquid guide away from the mist outlet tube **120** by insertion and fits the inner peripheral side of the end of the mist outlet tube **120** away from the mist outlet tube **120**, it just hinders the liquid from flowing out of the position in the liquid guide close to the support part **314**, thereby improving the sealing of the atomizer. As shown in FIG. **4**, a part of the base assembly **300** inserted into the mist outlet tube **120** may be a protrusion **312**, which may be tubular. In another one or more embodiments, the protrusion **312** may have other irregular shapes. In addition, it should be noted that since the atomizer and the aerosol generating apparatus are often portable in actual use, they are inevitably placed in various postures by the user. Thus, in many cases, the atomizer is not in the upright position as shown in FIG. **3**, but may be on its side or even upside down. However, when the atomizer is used by the user, it is usually placed in an upright position similar to that shown in FIG. **3**, because when the user suctioning the aerosol, the user tends to lower its head, rather than lift its head, to suction the aerosol. When the aerosol is suctioned by the user, a great amount of the liquid in the liquid guide is consumed to become atomized mist, mixed with gas to form the aerosol, and then discharged from the mist outlet **121**. As a result, liquid is constantly replenished from the liquid storage tank **140** to the liquid guide, so that the liquid flow on the liquid guide is the most intense during the use of the user, thereby easily causing the liquid to overflow the liquid guide. Thus, when the atomizer is at the upright position shown in FIG. **3** (that is, at the position, the mist outlet direction of the atomizer and the direction of gravity are approximately opposite), the end of the liquid guide close to the support part **314** is blocked, that is, the sealing effect of the atomizer may be greatly improved. The part of the base assembly **300** inserted into the liquid guide may fit the inner peripheral side of the liquid guide through an interference fit or an at least partial clearance fit. This is because the surface tension of the liquid is large, and even if there is an at least partial clearance fit, a blocking effect can be achieved. The fit method may be selected based on actual conditions.

[0039] As shown in FIG. **1**, in one or more embodiments, the base assembly **300** includes the protrusion **312** having an end extending toward the mist outlet tube **120** for being connected to an end of the liquid guide away from the mist outlet tube **120** by insertion, and a support part **314**. The protrusion **312** may include a soft rubber material, such as silica gel or rubber, to improve the sealing effect. However, the protrusion **312** may include a plastic material, a metal material, or the like, to improve the support for the atomization assembly **200**, and also have a certain sealing effect. The support part **314** surrounds an end of the protrusion **312** away from the mist outlet tube **120**, and the side edge of the support part **314** away from the protrusion **312** abuts against the inner wall of the liquid cup **100**, so that the support part **314**, the liquid cup **100**, the mist outlet tube **120**, and the atomization tube **210** form the liquid storage tank **140**. The support part **314** may be made of a soft rubber material same as the protrusion **312**, so that the sealing effect of the liquid storage tank **140** may be improved. The liquid in the liquid storage tank **140** is prevented from leaking out of the liquid storage tank **140** through the fit clearance between the support part **314** and the inner wall of the liquid cup **100**. The support part **314** surrounds the end of the protrusion **312** away from the mist outlet tube **120**, and it may be understood that the protrusion **312** extends through the

support part **314**. The protrusion **312** and the support part **314** may be integrated so that there is no fit clearance between the protrusion **312** and the support part **314**, thereby preventing liquid from leaking out of the liquid storage tank **140** through the fit clearance between the protrusion **312** and the support part **314**. However, the protrusion **312** and the support part **314** may also be provided separately in one or more embodiments, as described below.

[0040] As shown in FIG. **1**, in one or more embodiments, the liquid guide includes the first liquid guide **220** and the second liquid guide **230**. The first liquid guide **220** and the second guide **230** are both tubular in shape. The first liquid guide **220** may include a fiber material, a porous body, or the like. The fiber material may be a plant fiber, an animal fiber, a mineral fiber, a synthetic fiber, or the like. The material of the first liquid guide **220** may be cotton, non-woven fabric, glass fiber material, porous ceramic material, porous metal material, or the like. The material of the second liquid guide **230** may be a fiber material, a porous body, or the like. The fiber material may be a plant fiber, an animal fiber, a mineral fiber, a synthetic fiber, or the like. The material of the second liquid guide **230** may be cotton, non-woven cloth, glass fiber material, porous ceramic material, porous metal material, or the like. The atomization tube **210** is sleeved on the outer peripheral side of the first liquid guide **220**, the first liquid guide **220** is sleeved on the outer peripheral side of the second liquid guide **230**, and the heating part **240** is disposed in the second liquid guide **230**. In this way, it is necessary for the liquid to pass through the first liquid guide **220** to enter the second liquid guide **230** so as to be in contact with the heating part **240**, thereby extending the distance of the liquid path, increasing the blocking effect of the liquid guide on the liquid, and reducing the risk of the liquid breaking through the surface of the liquid guide (especially the second liquid guide **230**) to flow out. An end of the base assembly **300** close to the mist outlet tube **120** is connected to an end of the first liquid guide **220** away from the mist outlet tube **120** by insertion, and an end face of the base assembly **300** close to the mist outlet tube **120** at least partially abuts against an end face of the second liquid guide **230**. In this way, when the atomizer is used by the user, the lower end of the second liquid guide **230** in the gravity direction may be blocked, so that the liquid is prevented from breaking through the surface of the second liquid guide **230** at the position to leak out, and the sealing effect of the liquid is improved. Also, the second liquid guide may be reinforced to prevent the second liquid guide from falling off and deforming, and the service life of the second liquid guide is extended.

[0041] As shown in FIG. **1**, in one or more embodiments, the base assembly **300** further includes the first extension part **315**. The first extension part **315** has an end connected to one side of the support part **314** close to the housing **110**, and an opposite end extending away from the mist outlet tube **120**, which is equivalent to providing a skirt for the support part **314**. The outer peripheral side of the first extension part **315** abuts against the inner wall of the housing **110**, so that the contact area between the support part **314** and the inner wall of the liquid cup **100** is expanded. Thus, when the liquid penetrates into the fit clearance between the support part **314** and the liquid cup **100** and breaks through the support part **314**, the first extension part **315** also generates a sealing effect, thereby preventing leakage of the liquid and improving the sealing effect.

[0042] As shown in FIG. **2**, in one or more embodiments, the base assembly **300** further includes a reinforcing sheet **316**. The reinforcing sheet **316** is attached to one side of the support part **314** away from the liquid storage tank **140** to support the support part **314**. The reinforcing sheet **316** may have a hardness greater than the support part **314**, for example, the reinforcing sheet **316** may adopt a steel sheet or a hard plastic, and the support part **314** may adopt a soft rubber. The support part **314** may perform a good sealing effect, and the reinforcing sheet **316** prevents deformation of the support part **314**. The deformation of the support part **314** may cause the support part **314** to be unable to abut against the inner wall of the liquid cup **100**, thereby causing leakage. Therefore, the reinforcing sheet **316** indirectly improves the sealing effect.

[0043] As shown in FIGS. **2**, **6**, and **7**, in one or more embodiments, for example, as shown in FIG. **2**, the reinforcing sheet **316** surrounds an end of the protrusion **312** away from the mist outlet tube

120, and the protrusion **312** protrudes out of the side of the reinforcing sheet **316** away from the support part **314**. That is, as shown in FIGS. **2** and **6**, since the protrusion **312** protrudes out of the side of the reinforcing sheet **316** away from the support part **314**, the end of the protrusion **312** protruding out of the reinforcing sheet **316** may form a liquid collection ring **3121**. As shown in FIG. **2**, the bottom of the liquid collection ring **3121** is lower than the bottom of the reinforcing sheet **316**. In other words, the side of the liquid collection ring **3121** away from the second guide **230** is more protrusive with respect to the side of the reinforcing sheet **316** away from the second guide **230**. The liquid collection ring **3121** may prevent the condensate from flowing into the position where the reinforcing sheet **316** is located, so that the sealing effect is better, and leakage is effectively prevented. If the condensate droplets flow toward the base **320** along the inner side wall of the protrusion **312** during the operation of the atomizer, they will eventually converge onto the liquid collection ring **3121** and are not likely to flow further outwardly in the radial direction of the protrusion **312** onto the reinforcing sheet **316**. The condensate is thus controllable. Specifically, when the condensate reaches a certain amount, the droplets will fall. Since the droplets are hung on the liquid collection ring **3121**, the position of the falling is substantially at the projection of the liquid collection ring **3121** on the base **320**. A leakage passage or a structure such as a liquid absorbing member **500** mentioned below may be provided at the position to discharge the condensed liquid out of the atomizer or absorb the condensed liquid into the liquid absorbing member **500**, so as to prevent the condensed liquid from flowing to other devices to cause damage. For example, a common danger of condensate is that it will flow to the breath detector in the aerosol generating apparatus to damage the breath detector.

[0044] As shown in FIG. **7**, in one or more embodiments, the reinforcing sheet **316** surrounds an end of the protrusion **312** away from the mist outlet tube **120**, and the protrusion **312** and the reinforcing sheet **316** are integrated. As shown in FIG. **8**, the protrusion **312** and the reinforcing sheet **316** are integrated to form a stronger support structure, that is, an end of the atomization assembly **200** close to the base **320** may be fixed, and the support part **314** may be fixed. Each of the protrusion **312** and the reinforcing sheet **316** may adopt a hard material such as plastic or metal (including alloy) to provide a good supporting effect.

[0045] As shown in FIG. **7**, in one or more embodiments, the side of the support part **314** facing the liquid storage tank **140** is provided with a gas release groove **314a**. The gas release groove **314a** extends from the connection position between the protrusion **312** and the support part **314** to the liquid storage tank **140** in the radial direction of the protrusion **312** away from the protrusion **312**. This is equivalent to connecting the fit clearance between the protrusion **312** and the liquid guide with the liquid storage tank **140**, thereby forming a gas passage. When the amount of liquid in the liquid storage tank **140** decreases, the gas may be replenished to the liquid storage tank **140** through the gas passage to balance the gas pressure, so that the liquid in the storage tank **140** is prevented from being unable to flow to the heating part **240** due to excessive negative pressure. For details, please refer to the description of the first gap **312c** in the following.

[0046] As shown in FIG. **2**, in one or more embodiments, an end of the atomization tube **210** facing the base assembly **300** is at least partially spaced apart from the support part **314** to form a liquid inlet **211** for connecting the liquid guide with the liquid storage tank **140**. Since the support part **314** is located at the bottom of the liquid storage tank **140**, the liquid inlet **211** is provided here to fully utilize the liquid, thereby preventing waste caused by residual liquid in the liquid storage tank **140**. Also in another one or more embodiments, as shown in FIG. **1**, the heating part **240** is spaced apart from the liquid inlet **211** in the axial direction of the atomization tube **210**, and the heating part **240** is higher than the liquid inlet **211** when the atomizer is used by the user. This means that in order for the liquid to reach the heating part **240**, it is first necessary for the liquid to climb up to the heating part **240** in the liquid guide against the action of gravity. This weakens the pushing action of a part of the liquid in the liquid storage tank **140** on the liquid in the liquid guide, and prevents leakage caused by the liquid breaking through the surface of the liquid guide due to

excessive pressure in the liquid guide. As shown in FIG. 1, the protrusion 312 also blocks the inner peripheral side of the liquid guide, thereby further improving the sealing effect.

[0047] As shown in FIG. 1, in one or more embodiments, the height of the inlet 211 starting from the support part 314 in the axial direction of the atomization tube 210 is a. The depth of the protrusion 312 extending into the liquid guide is b. a and b satisfy the constraint $1.5a \leq b \leq 10a$. When $b < 1.5a$, the protrusion 312 has a poor effect on supporting the atomization assembly 200, and the atomization assembly 200 is likely to shake, thereby causing leakage. When $b > 10a$, the heating part 240 is raised too high (because the heating part 240 cannot be shielded by the protrusion 312, but should be exposed in the atomizing passage formed by the protrusion 312, the atomization assembly 200, and the mist outlet tube 120 which are connected) so that the liquid hardly reaches the heating part 240, thereby causing the heating part 240 to dry burn and damage the heating part 240. However, when b is in the above range, the above problems may be overcome.

[0048] As shown in FIGS. 1 and 10, in one or more embodiments, the atomizer further includes a first electrode 410 and a second electrode 420. The base assembly 300 includes a base 320 and a sealing member 310. The base 320 may be inserted into the installation chamber 130 from the cup mouth, so that the base assembly 300 is connected to the liquid cup 100 by insertion. The sealing member 310 includes a protrusion 312 and a support part 314, and is disposed on the side of the base 320 facing the atomization assembly 200. The first electrode 410 and the second electrode 420 are spaced apart from each other and both extend through the base 320. The protrusion 312 is provided with a lead channel extending through the protrusion 312 in the axial direction of the protrusion 312, and the lead channel may include a first lead channel 312a and a second lead channel 312b. The first electrode 410 may be electrically connected to the heating part 240 through the lead in the first lead channel 312a. The second electrode 420 may be electrically connected to the heating part 240 through the lead in the second lead channel 312b. One of the first electrode 410 and the second electrode 420 may be connected to the positive electrode of the heating part 240, and the other may be connected to the negative electrode of the heating part 240. The first electrode 410 and the second electrode 420 are actually structures of the atomizer electrically connected to the body of the aerosol generating apparatus. Generally, the body extends two contacts (or pins). When the atomizer is connected to the body, the two contacts are in contact with the first electrode 410 and the second electrode 420, respectively, to implement electrical connection. In a general aerosol generating apparatus, the leads of the positive electrode and negative electrode of the heating part 240 are suspended in the mist outlet channel, and are likely to be in contact with each other to cause a short circuit. In one or more embodiments of the present disclosure, the leads of the two electrodes of the heating part 240 are provided in the first lead channel 312a and the second lead channel 312b, respectively, so that it is ensured that the leads are spaced apart from each other and it is not likely to generate a short circuit. In addition, as shown in FIGS. 1 and 10, the first electrode 410 may be an electrode nail, and the second electrode 420 may be an electrode ring. The electrode nail and the electrode ring are separated by an insulating silica gel member 430 to form separate electrodes. Of course, the first electrode 410 and the second electrode 420 may be in the form of general electrodes as long as the first electrode 410 and the second electrode 420 are spaced apart from each other.

[0049] As shown in FIG. 6, in one or more embodiments, a glue groove 311 is provided on the side of the sealing member 310 away from the atomization assembly 200. The first lead channel 312a and the second lead channel 312b both extend through the bottom of the glue groove 311. An end of each of the first lead channel 312a and the second lead channel 312b is undoubtedly connected with the liquid guide, since the heating part 240 is provided in the liquid guide. Thus, the liquid may leak along the first lead channel 312a or the second lead channel 312b. Therefore, the glue groove 311 is provided so that a sealant may be filled in the glue groove 311 to block an end of each of the first lead channel 312a and the second lead channel 312b away from the heating part 240 to prevent liquid leakage.

[0050] As shown in FIG. 3, in one or more embodiments, the atomizer further includes a liquid absorbing member **500**. The liquid absorbing member **500** is disposed on the side of the base **320** facing the protrusion **312** and opposite to the protrusion **312**. The liquid absorbing member **500** may adopt a material having a strong adsorption capacity for the liquid, for example, a material similar to the liquid guide. In this way, the liquid falling from the protrusion **312** may be collected into the liquid absorbing member **500**, so that the damage to the aerosol generating apparatus due to condensed liquid flowing to the electronic apparatus is prevented.

[0051] As shown in FIG. 1, in one or more embodiments, the base assembly **300** includes a sealing member **310** that includes a protrusion **312**. The protrusion **312** is disposed facing the atomization assembly **200**. As shown in FIG. 1, the protrusion **312** is inserted into an end of the atomization assembly **200** away from the mist outlet tube **120**. However, in another one or more embodiments, the protrusion **312** is sleeved on the outer peripheral side of the atomization assembly **200**. Both arrangements allow the base assembly **300** to be connected to an end of the atomization assembly **200** away from the mist outlet tube **120** by insertion. The former may block the inner peripheral side of the liquid guide, and the latter may block the fit clearance between the atomization assembly **200** and the base assembly **300**. Both may improve sealing.

[0052] As shown in FIGS. 1 and 10, in one or more embodiments, the atomizer further includes a liquid injection cover **600**. The housing **110** is provided with a liquid injection hole connected with the installation chamber **130**, and the liquid injection cover **600** covers the liquid injection hole. Thus, when there is insufficient liquid in the liquid storage tank **140**, the liquid injection cover **600** may be opened to fill the liquid storage tank **140**. This allows the atomizer to be reused. In another one or more embodiments, the atomizer may, of course, not be provided with a liquid injection cover **600**, and is a disposable article for ease of use by the user.

[0053] In one or more embodiments, the atomization assembly **200** and the base assembly **300** may be first obtained during assembly. The atomizer is in the state shown in FIG. 12, as can be seen that the atomizer includes three modules, that is, the liquid cup **100**, the atomization assembly **200** and the base assembly **300**. An end of the atomization assembly **200** is connected to the mist outlet tube **120** by insertion. The atomization assembly **200** is connected to the mist outlet tube **120** by insertion. The base assembly **300** is connected to the housing **110** by insertion so as to cover the cup mouth. After the base assembly **300** is connected to the housing **110** by insertion, the base assembly **300** is also naturally connected to the atomization assembly **200** by insertion. It can be seen that due to the modular design, the assembly of the atomizer may be made more convenient.

[0054] In another one or more embodiments, during assembly, an end of the atomization tube **210** may also be inserted into an end of the mist outlet tube **120** away from the mist outlet **121** to form a first assembly **710**, and after the base assembly **300** is assembled, the liquid guide is connected to the base assembly **300** by insertion so as to form a second assembly **720**. The atomizer is in the state shown in FIG. 13. The second assembly **720** is inserted into the first assembly **710** from the cup mouth in the axial direction of the atomization tube **210** to complete the assembly of the atomizer. That is, after the second assembly **720** is inserted into the first assembly **710** from the cup mouth, the liquid guide is assembled to the atomization tube **210** while the base assembly **300** is connected to the liquid cup **100** by insertion, so that the assembly is completed. It can be seen that the atomizer may be modularized into the first assembly **710** and the second assembly **720**, also with a high degree of modularity, making the assembly of the atomizer more convenient.

[0055] As shown in FIG. 1, according to one or more embodiments of the present disclosure, the atomizer includes the liquid cup **100**, the atomization assembly **200**, and the base assembly **300**. An end of the liquid cup **100** is provided with a cup mouth, and the other end of the liquid cup **100** is provided with a mist outlet **121**. The liquid storage tank **140** is provided in the liquid cup **100**. The atomization assembly **200** includes the atomization tube **210**, the liquid guide, and the heating part **240**. The heating part **240** is provided in the liquid guide, the atomization tube **210** is connected with the mist outlet **121** and is sleeved on the outer peripheral side of the liquid guide. The

atomization tube **210** is provided with the liquid inlet **211** so that the liquid guide is connected with the liquid storage tank **140**. The base assembly **300** includes the base **320** and the sealing member **310**. The base **320** covers the cup mouth. The sealing member **310** is installed at the base **320** and extends along a side close to the mist outlet **121** for being connected to an end of the liquid guide away from the mist outlet **121** by insertion. The sealing member **310**, the atomization tube **210** and the mist outlet **121** are connected to form a mist passage. The sealing member **310** fits at least a portion of the inner wall of the liquid guide to block the radially inner side of the liquid guide. The liquid is guided from the liquid storage tank **140** through capillary action. Therefore, the liquid guide is filled with liquid under normal circumstances. However, the distribution of the liquid in the liquid guide is not completely uniform. The liquid tends to accumulate towards the direction in which the gravity is directed. While the atomizer is generally approximately in the upright position as shown in FIG. **1**, that is, the gas flow direction in the mist outlet passage is approximately in the direction of anti-gravity. Thus, the liquid generally accumulates at the end of the liquid guide facing the base **320**. Therefore, the sealing member **310** fits with at least part of the inner wall the liquid guide to block the radially inner side of the liquid guide, so that the sealing may be improved and the liquid leakage is prevented. In addition, the sealing member **310** may abut against or be spaced apart from the inner wall of the liquid guide. Since the liquid has a great surface tension, even if the sealing member **310** is spaced apart from the inner wall of the liquid guide, the liquid may be greatly hindered from flowing inwardly along the radial direction of the liquid guide.

[0056] As shown in FIG. **1**, in one or more embodiments, the sealing member **310** includes the protrusion **312** and the support part **314**. The support part **314** abuts against the inner wall of the liquid cup **100**, so that the support part **314**, the liquid cup **100** and the atomization tube **210** form the liquid storage tank **140**. An end of the protrusion **312** extends through the support part **314**, and the other end of the protrusion **312** extends toward the mist outlet **121** so as to be connected to an end of the liquid guide away from the mist outlet **121** by insertion. Thus, the surface of the end of the sealing member **310** facing the base **320** is located in the liquid storage tank **140**, and there is no need for blocking and there is no leakage therefrom. The support part **314** also improves the sealing effect of the liquid storage tank **140** and prevents liquid from leaking directly from the fit clearance between the base assembly **300** and the liquid cup **100**.

[0057] As shown in FIG. **11**, in one or more embodiments, multiple abutments **212** protruding axially along the atomization tube **210** are provided on an end of the atomization tube **210** close to the support part **314**, and the abutments **212** abut the support part **314**. In this way, the support part **314** may support an end of the atomization tube **210** close to the support part **314**, thereby improving the fixing effect of the atomization assembly **200**. The multiple abutments **212** are arranged at intervals to form multiple liquid inlets **211** arranged at equal intervals in the circumferential direction of the atomization tube **210**. Thus, the liquid may enter the liquid guide in all directions, thereby improving the liquid supply capacity of the aerosol generating apparatus. As shown in FIG. **2**, in another one or more embodiments, an end of the atomization tube **210** close to the support part **314** is spaced apart from the support part **314** to form the liquid inlet **211**. This has a simple structure and simplifies the manufacture of the atomization tube **210**. Also, the length of the liquid inlet **211** in the circumferential direction of the atomization tube **210** is maximized (i.e., equal to the circumferential length of the atomization tube **210**), thereby ensuring that the liquid may enter the liquid guide from all directions and improving the liquid supply capacity.

[0058] As shown in FIG. **1**, in one or more embodiments, the sealing member **310** further includes the first extension part **315**. The first extension part **315** is provided on the peripheral side of the support part **314** and extends away from the liquid storage tank **140**. The outer peripheral side of the first extension part **315** abuts against the inner wall of the liquid cup **100**. In this way, when the liquid breaks through the fit clearance between the support part **314** and the inner wall of the liquid cup **100**, it is blocked by the first extension part **315**, thereby improving the sealing effect of the liquid storage tank **140**.

[0059] In one or more embodiments, the protrusion **312** extends through the reinforcing sheet **316** and protrudes out of a side of the reinforcing sheet **316** away from the support part **314**. That is, as shown in FIGS. **2** and **6**, since the protrusion **312** protrudes out of the side of the reinforcing sheet **316** away from the support part **314**, the side of the protrusion **312** protruding out of the reinforcing sheet **316** forms a liquid collection ring **3121**. If during the operation of the atomizer the condensate droplets flow toward the base **320** along the inner side wall of the protrusion **312**, they will eventually converge onto the liquid collection ring **3121** and are not likely to further flow outwardly in the radial direction of the protrusion **312** onto the reinforcing sheet **316**. The condensate is thus controllable. Specifically, when the condensate reaches a certain amount, the droplets will fall. Since the droplets are hung on the liquid collection ring **3121**, the position of the falling is substantially at the projection of the liquid collection ring **3121** on the base **320**. A leakage passage or a structure such as the liquid absorbing member **500** mentioned below may be provided to discharge the condensed liquid from the atomizer or absorb the condensed liquid into the liquid absorbing member **500**, so as to prevent the condensed liquid from flowing to other devices. For example, a common danger of condensate is that it will flow to the breath detector in the aerosol generating apparatus and the head is damaged.

[0060] As shown in FIG. **1**, in one or more embodiments, the base **320** includes a cover **321** and a second extension part **322**. The cover **321** covers the opening of the liquid cup **100**, and the second extension part **322** surrounds the cover **321** and extends toward the support part **314**. The outer peripheral side of the second extension part **322** abuts against the first extension part **315**, so that the first extension part **315** may better abut against the inner wall of the liquid cup **100**, thereby improving the sealing effect. The end face of the second extension part **322** away from the cover **321** may abut against the reinforcing sheet **316**. In this way, the reinforcing sheet **316** may be supported to improve the fixing effect of the reinforcing sheet **316**.

[0061] As shown in FIG. **4**, in one or more embodiments, a gas release groove **314a** is provided on the side of the support part **314** facing the liquid storage tank **140**. The gas release groove **314a** extends away from the protrusion **312** in the radial direction of the protrusion **312** from the connection position between the protrusion **312** and the support part **314**. The gas release groove **314a** is connected with the liquid storage tank **140**. In this way, the gas release groove **314a** may supply gas to the liquid storage tank **140** through the second gap between the liquid guide and the protrusion **312**, so that the liquid is prevented from failing to be transported to the heating part **240** through the liquid guide due to the negative pressure caused by the consumption of the liquid in the liquid storage tank **140**.

[0062] As shown in FIG. **1**, in one or more embodiments, the liquid guide includes the first liquid guide **220** and the second liquid guide **230**. The atomization tube **210** is sleeved on the outer peripheral side of the first liquid guide **220**. The first liquid guide **220** is sleeved on the outer peripheral side of the second guide **230**. The heating part **240** is disposed in the second liquid guide **230**. The radially outer peripheral side of the protrusion **312** abuts against the radially inner peripheral side of the first liquid guide **220**. The end face of the protrusion **312** away from the support part **314** abuts against the end face of the second liquid guide **230**. In this way, the radially inner side of the first liquid guide **220** and the end face of the second guide **230** facing the protrusion **312** are both blocked, thereby improving the sealing effect of the liquid.

[0063] According to the atomizer provided in one or more embodiments of the present invention, the mist outlet tube is provided on the liquid cup, a first end of the atomization assembly is connected to an end of the mist outlet tube away from the mist outlet, and a second end of the atomization assembly is connected to the protrusion of the base assembly, so that the atomizer may be assembled more conveniently. Specifically, when the mist outlet tube and the housing are integrated, the problem of connecting the mist outlet tube to the housing does not need to be taken into account when assembling, thereby making the assembly easier. When the mist outlet tube and the housing are separated, since the atomization assembly is not disposed in the mist outlet tube,

but is connected to the end of the mist outlet tube away from the mist outlet, the atomization tube does not need to extend to the base assembly independently, but may be extended to the base assembly through the atomization assembly. Thus, the length of the atomization tube is small and the assembly difficulty is small. In addition, the atomization core is connected to each of the mist outlet tube and the base assembly by insertion, so that the atomization core does not need operations such as rotation (if a threaded connection is used, rotation is usually required) when being assembled to the mist outlet tube, and the atomization core only needs to be pushed toward the mist outlet tube in the insertion direction. Similarly, the base assembly may also only need to be pushed toward the atomization core in the insertion direction so as to be connected, and the assembly process is more convenient. In summary, according to the aerosol generating apparatus provided in one or more embodiments of the present disclosure, the assembly of the aerosol generating apparatus may be more convenient.

[0064] In the above-mentioned embodiments, the description of each embodiment has its own emphasis. For a part not described in detail in a certain embodiment, the related description in other embodiments may be referred to.

[0065] In the description of the present disclosure, the terms “first” and “second” are used for descriptive purposes only and are not to be construed as indicating or implying relative importance or implying the number of indicated technical features. Thus, a feature defined as “first” or “second” may explicitly or implicitly include one or more features.

[0066] Some embodiments of the present disclosure have been described in detail above. The description of the above embodiments merely aims to help to understand the present disclosure. Many modifications or equivalent substitutions with respect to the embodiments may occur to those of ordinary skill in the art based on the present disclosure. Thus, these modifications or equivalent substitutions shall fall within the scope of the present disclosure.

Claims

1. An atomizer, comprising: a liquid cup having an end provided with a mist outlet; an atomization assembly comprising an atomization tube, a liquid guide part, and a heating part disposed inside the liquid guide part, the atomization tube being sleeved on an outer peripheral side of the liquid guide part, the atomization tube having a first end connected with the mist outlet and a second end provided with a liquid inlet; and a sealing member comprising a support part and a protrusion, wherein the support part abuts against an inner wall of the liquid cup; the support part, the liquid cup and the atomization tube form a liquid storage tank connected with the liquid guide part through the liquid inlet; the protrusion extends into the liquid guide part from an end of the liquid guide part away from the mist outlet, so that the protrusion, the atomization tube and the mist outlet form a mist outlet passage; and the support part is provided with at least one ventilation passage for connecting the liquid storage tank with atmosphere.

2. The atomizer according to claim 1, wherein the protrusion fits with at least part of an inner wall of the liquid guide part to form a first gap; the liquid guide part is at least partially spaced apart from the support part to form a second gap; and the first gap is connected with the second gap to form a gas inlet passage connecting the mist outlet passage with the liquid storage tank.

3. The atomizer according to claim 2, wherein an end face of the liquid guide part facing the support part abuts against the support part, and a side of the support part facing the liquid guide part is provided with one or more gas release grooves of the ventilation passage to form the second gap.

4. The atomizer according to claim 3, wherein the atomization tube has a projection on the support part, and each of the gas release grooves extends and protrudes from the projection in a radial direction of the protrusion.

5. The atomizer according to claim 3, wherein the one or more gas release grooves comprise a

plurality of gas release grooves arranged at equal intervals in a circumferential direction of the protrusion.

6. The atomizer according to claim 2, wherein the liquid guide part comprises a first liquid guide and a second liquid guide; the atomization tube is sleeved on an outer peripheral side of the first liquid guide; the first liquid guide is sleeved on an outer peripheral side of the second liquid guide; the heating part is disposed in the second liquid guide; an outer peripheral side of the protrusion fits with an inner peripheral side of the first liquid guide to form the first gap; and an end face of the protrusion away from the support part fits with an end face of the second liquid guide to form a third gap connected with the first gap.

7. The atomizer according to claim 2, wherein the sealing member further comprises a first extension part surrounding the support part and extending away from the liquid storage tank, an outer peripheral side of the first extension part abutting against the inner wall of the liquid cup.

8. The atomizer according to claim 7, wherein the outer peripheral side of the first extension part is provided with a convex rib extending along a circumferential direction of the first extension part and surrounding the first extension part.

9. The atomizer according to claim 2, wherein the sealing member further comprises a reinforcing sheet attached to a side of the support part away from the liquid storage tank to support the support part.

10. The atomizer according to claim 9, wherein the reinforcing sheet surrounds an end of the protrusion away from the mist outlet; and the protrusion protrudes from a side of the reinforcing sheet away from the support part.

11. The atomizer according to claim 9, wherein the reinforcing sheet surrounds an end of the protrusion away from the mist outlet; and the protrusion is integrally formed with the reinforcing sheet and comprises a same material as the reinforcing sheet.

12. The atomizer according to claim 2, wherein the liquid guide part comprises a first liquid guide and a second liquid guide; the atomization tube is sleeved on an outer peripheral side of the first liquid guide; the first liquid guide is sleeved on an outer peripheral side of the second liquid guide; the heating part is disposed in the second liquid guide; an outer peripheral side of the protrusion fits with an inner peripheral side of the first liquid guide to form the first gap; an end face of the protrusion away from the support part fits with an end face of the second liquid guide to form a third gap; and the sealing member further comprises a heat insulation sleeve sleeved on an end of the protrusion close to the second liquid guide.

13. An aerosol generating apparatus, comprising a body and an atomizer connected to the body, wherein the atomizer comprises: a liquid cup having an end provided with a mist outlet; an atomization assembly comprising an atomization tube, a liquid guide part, and a heating part disposed inside the liquid guide part, the atomization tube being sleeved on an outer peripheral side of the liquid guide part, the atomization tube having a first end connected with the mist outlet and a second end provided with a liquid inlet; and a sealing member comprising a support part and a protrusion, wherein the support part abuts against an inner wall of the liquid cup; the support part, the liquid cup and the atomization tube form a liquid storage tank connected with the liquid guide part through the liquid inlet; the protrusion extends into the liquid guide part from an end of the liquid guide part away from the mist outlet, so that the protrusion, the atomization tube and the mist outlet form a mist outlet passage; and the support part is provided with at least one ventilation passage for connecting the liquid storage tank with atmosphere.

14. The aerosol generating apparatus according to claim 13, wherein the protrusion fits with at least part of an inner wall of the liquid guide part to form a first gap; the liquid guide part is at least partially spaced apart from the support part to form a second gap; and the first gap is connected with the second gap to form a gas inlet passage connecting the mist outlet passage with the liquid storage tank.

15. The aerosol generating apparatus according to claim 14, wherein an end face of the liquid guide

part facing the support part abuts against the support part, and a side of the support part facing the liquid guide part is provided with one or more gas release grooves of the ventilation passage to form the second gap.

16. The aerosol generating apparatus according to claim 15, wherein the atomization tube has a projection on the support part, and each of the gas release grooves extends and protrudes from the projection in a radial direction of the protrusion.

17. The aerosol generating apparatus according to claim 15, wherein the one or more gas release grooves comprise a plurality of gas release grooves arranged at equal intervals in a circumferential direction of the protrusion.

18. The aerosol generating apparatus according to claim 14, wherein the liquid guide part comprises a first liquid guide and a second liquid guide; the atomization tube is sleeved on an outer peripheral side of the first liquid guide; the first liquid guide is sleeved on an outer peripheral side of the second liquid guide; the heating part is disposed in the second liquid guide; an outer peripheral side of the protrusion fits with an inner peripheral side of the first liquid guide to form the first gap; and an end face of the protrusion away from the support part fits with an end face of the second liquid guide to form a third gap connected with the first gap.

19. The aerosol generating apparatus according to claim 14, wherein the sealing member further comprises a first extension part surrounding the support part and extending away from the liquid storage tank, an outer peripheral side of the first extension part abutting against the inner wall of the liquid cup.

20. The aerosol generating apparatus according to claim 19, wherein the outer peripheral side of the first extension part is provided with a convex rib extending along a circumferential direction of the first extension part and surrounding the first extension part.
